

Route 1: NL

Prompt

Solve the algorithmic problem step by step in natural language.

Return JSON with keys:
simulation, answer.
constraint: no code

Example Q: 123 + 456

```
{  
  "simulation":  
    "3+6=9, 2+5=7,  
    1+4=5",  
  "answer": "579"  
}
```

Final answer

579

Route 2: Code + sim.

Prompt

Solve the algorithmic problem. Return code and a NL simulation.

Return JSON with keys:
code, simulation, answer.
constraint: same JSON schema

Example Q: 123 + 456

```
{  
  "code":  
    "def solution():  
      return 123+456",  
  "simulation": "adds",  
  "answer": "579"  
}
```

Final answer

579

Route 3: Code + Python

Procedure

1. Extract solution() from the Route 2 response.
2. Run in sandboxed Python (5 s).
3. Compare to gold.
no additional LLM call

Code from Route 2

```
def solution():  
    return 123 + 456  
  
>>> solution()  
579
```

Final answer

579